

ZVERINOGOLOVSKOYE, Russian FederationWMO: **287560**Lat: **54.467N**Lon: **64.867E**Elev: **117**StdP: **99.93**Time Zone: **5.00 (E05)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-29.4	-35.4	0.1	-32.4	-32.4	0.2	-29.3	10.0	-17.2	8.5	-16.5	1.5	180	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.8	20.6	30.9	19.9	28.9	19.1	22.5	29.4	21.2	28.4	20.1	26.7	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.0	26.5	18.8	13.8	24.2	17.7	12.9	23.4	66.6	29.3	62.2	28.7	58.2	26.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.4	-36.2	35.7	3.7	1.7	-38.8	36.9	-41.0	37.9	-43.0	38.8	-45.7	40.1		
(5)			-36.3	23.8	3.6	2.0	-38.9	25.3	-41.0	26.5	-43.1	27.6	-45.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-17.8	-15.2	-7.0	5.1	13.7	18.7	20.3	18.1	12.3	4.0	-4.9	-13.9		
	DBStd	14.68	6.37	7.15	5.79	5.88	5.02	4.77	3.97	4.42	5.09	5.11	6.36	7.71		
	HDD10.0	3706	862	705	528	167	22	3	0	1	33	194	448	742		
	HDD18.3	5868	1121	938	787	396	158	53	24	61	189	444	698	1000		
	CDD10.0	1104	0	0	0	22	136	264	321	251	103	9	0	0		
	CDD18.3	227	0	0	0	1	14	63	87	53	9	0	0	0		
	CDH23.3	2629	0	0	0	12	226	786	954	544	107	0	0	0		
	CDH26.7	969	0	0	0	2	63	299	382	192	31	0	0	0		
(14) Wind	WSAvg	2.7	2.7	2.9	3.2	3.2	3.0	2.6	2.5	2.2	2.3	2.6	2.8	2.9		
(15) Precipitation	PrecAvg	364	17	14	18	21	34	43	56	47	30	29	27	24		
	PrecMax	578	39	42	42	48	115	128	133	119	57	75	51	57		
	PrecMin	189	2	2	0	2	2	5	4	18	1	2	5	3		
	PrecStd	107	11	9	11	12	27	34	33	26	17	16	13	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	2.0	9.0	24.9	31.4	34.3	36.0	33.8	30.4	20.3	8.9	2.0		
		MCWB	-2.0	0.3	5.6	14.0	18.6	20.4	20.9	22.5	17.4	12.0	6.3	0.3		
	2%	DB	-3.3	0.1	3.7	20.8	28.4	32.2	32.9	31.1	26.3	17.0	5.9	0.0		
		MCWB	-4.0	-0.8	1.9	11.9	16.7	19.9	20.8	20.6	15.7	10.6	4.5	-0.8		
	5%	DB	-5.5	-2.0	2.1	17.8	25.9	30.3	30.7	28.7	23.5	14.4	4.2	-1.8		
		MCWB	-6.2	-2.9	0.7	10.5	16.1	19.5	20.5	19.6	15.1	9.2	2.8	-2.6		
	10%	DB	-8.0	-4.8	1.0	14.8	23.5	28.0	28.6	26.4	20.7	11.7	2.4	-3.6		
		MCWB	-8.6	-5.6	-0.2	9.0	14.8	18.5	19.7	18.5	13.9	7.8	1.2	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	0.6	6.1	14.6	20.1	23.7	23.8	24.2	19.7	12.9	6.7	0.5		
		MCDB	-1.3	1.6	9.0	23.1	27.6	31.0	30.1	31.4	27.9	18.4	8.2	1.5		
	2%	WB	-3.9	-0.8	1.9	12.5	18.0	21.5	22.3	22.3	17.0	11.2	4.6	-0.9		
		MCDB	-3.3	0.2	3.2	19.4	25.5	29.5	29.6	28.7	23.4	16.1	5.9	0.0		
	5%	WB	-6.1	-2.8	0.9	10.9	16.7	20.1	21.4	20.4	15.7	9.6	2.9	-2.6		
		MCDB	-5.5	-2.1	1.9	17.1	24.8	27.8	28.5	26.5	22.3	13.8	4.1	-1.8		
	10%	WB	-8.6	-5.6	-0.1	9.3	15.3	19.0	20.5	19.0	14.2	8.1	1.3	-4.4		
		MCDB	-8.0	-4.9	0.9	14.6	22.5	26.8	26.9	24.6	20.1	11.2	2.3	-3.6		
(35) Mean Daily Temperature Range	MDBR		7.6	8.7	8.7	10.3	12.6	12.2	11.5	11.6	10.8	8.8	6.4	7.5		
	5% DB	MCDBR	8.9	9.0	7.4	15.2	16.5	15.5	14.4	14.8	14.4	13.2	6.5	7.7		
		MCWBR	8.6	8.2	6.1	8.1	7.8	7.0	5.5	6.7	7.1	8.0	5.2	7.1		
	5% WB	MCDBR	8.9	8.8	7.1	14.4	14.9	14.1	12.9	13.3	13.1	12.4	6.1	7.5		
		MCWBR	8.6	8.1	6.1	8.1	7.8	7.6	6.3	7.4	7.3	8.0	5.3	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.322	0.375	0.411	0.411	0.408	0.426	0.440	0.385	0.359	0.320	0.290		
	taud		2.058	2.015	1.955	2.124	2.217	2.300	2.247	2.179	2.287	2.243	2.157	2.041		
	Ebn at Noon		618	729	771	808	833	840	814	773	771	692	580	520		
	Edh at Noon		70	104	139	136	131	122	127	128	101	83	64	58		